

ABDOULAYE ALHASSANE DIALLO

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EDUCATION

Columbia University: B.A. in Computer Science, New York, NY

Expected May 2028

Jack Kent Cooke Scholar (national merit scholarship, <1% selection rate)

Coursework: Data Structures & Algorithms, CS Theory, Advanced Programming (C/C++), Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Node.js, Express.js, FastAPI, Qt, Tailwind CSS

Data & AI: PostgreSQL, SQLite, REST APIs, RAG pipelines, ChromaDB, sentence-transformers, LLM APIs (Groq, IBM Watson)

Infrastructure: Nginx, Cloudflare (DNS/SSL), Vercel, GitHub Actions (CI/CD), Linux, Git

Spoken Languages: English, French, Arabic, Fula

EXPERIENCE

Software Engineer & Tech Lead: Heuvia (heuvia.com), Remote

2025 – Present

- Lead a 5-person team (coordinating 3 developers) building a multi-tenant Learning Management System for colleges in Guinea, used by students, professors, and administrators
- Designed the system architecture: multi-tenant subdomain routing (Nginx + Cloudflare), a 4-role permission system, and curriculum-based enrollment on a React / React Native / Node.js / PostgreSQL stack
- Ship across web and mobile with production deployment, code review, and task coordination across the team

IRT Support Specialist: Research Foundation of CUNY, New York, NY

Jun – Aug 2025

- Resolved 200+ weekly Microsoft 365 / Active Directory requests (provisioning, MFA, access control) with a 99% resolution rate; deployed and configured 100+ Windows machines across campus systems

IT Manager: Guinea High Tech, Conakry, Guinea

2015 – 2022

- Scaled technical operations from 1 to 10+ team members, delivering end-to-end IT solutions to 250+ businesses; automated system deployment and standardized configurations across client environments

Web Development Intern: Identix Guinée, Conakry, Guinea

Jul – Nov 2018

- Developed web application features and contributed to relational database schema design using Laravel (PHP) and Bootstrap

PROJECTS

The Unofficial Guide: RAG Question-Answering System | Python, ChromaDB, sentence-transformers, Groq, Gradio github.com/Kienda

- Built an end-to-end retrieval-augmented generation system answering questions about Columbia CS professor reviews: collected 376 reviews via API, designed a per-review chunking strategy (394 chunks), and indexed embeddings (all-MiniLM-L6-v2) in ChromaDB
- Engineered grounded generation with source citations and out-of-scope refusal; evaluated retrieval quality across test queries and documented failure-case analysis with proposed metadata-filtering fixes

StudentSuccess: Academic Guidance Platform | Node.js, Express.js, PostgreSQL, JWT github.com/Kienda

- Built a full-stack platform with role-based authentication (JWT, bcrypt), a relational PostgreSQL schema, and server-rendered personalized student dashboards

College Navigator: Desktop Academic Tracker (Honors Project) | C++, Qt, SQLite

- Developed a cross-platform C++/Qt desktop app tracking academic milestones with SQLite persistence and an object-oriented architecture separating UI, logic, and data layers

Personal Portfolio & CI/CD Pipeline | Next.js, TypeScript, Tailwind, Vercel, Cloudflare abdoulayediallo.com

- Designed and shipped a production site with a serverless contact form (Resend API), GitHub-to-Vercel CI/CD, and custom domain/DNS/SSL via Cloudflare